

# Week 4 – Model Validation Report: BSM Error Metrics

**\*\*Run date\*\*:** 20260528 | **\*\*Pipeline version\*\*:** v1.0

## Methodology

The BSM analytical closed-form prices are treated as *model predictions*.

Monte Carlo (MC) simulation prices (N=10,000 paths, GBM under risk-neutral measure, seed=42) serve as the independent benchmark ("actual prices").

MAE and RMSE measure how closely the BSM formula approximates the MC benchmark across different market regimes.

Parameters are derived from JPM historical market data (2018–2024):

- \*\*S\*\***: JPM daily close price
- \*\*σ\*\***: 20-day rolling historical annualised volatility
- \*\*r\*\***: US 10-year Treasury yield (DGS10)
- \*\*q\*\***: trailing-twelve-month dividend yield
- \*\*VIX regime\*\***: Low < 20.0, Medium 20.0–30.0, High ≥ 30.0

Evaluation grid: 3 maturities × 3 moneyness levels × 2 option types, sampled monthly → **\*\*1,008 pricing observations\*\*** over **\*\*56 dates\*\***.

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## 1. Overall Error Metrics

| Metric             | Value    |
|--------------------|----------|
| MAE (overall)      | 0.112135 |
| RMSE (overall)     | 0.174916 |
| Max                | BSM – MC |
| Total observations | 1,008    |

These values quantify the numerical convergence gap between the BSM analytical formula and the MC simulation benchmark. Both are generated under identical GBM assumptions, so deviations arise purely from MC sampling variance (which decreases as  $\propto 1/\sqrt{N}$ ).

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## 2. Error by VIX Regime (Failure Mode Analysis)

| Regime             | MAE      | RMSE     | n   |
|--------------------|----------|----------|-----|
| Low (VIX < 20.0)   | 0.087432 | 0.128842 | 630 |
| Medium (20.0–30.0) | 0.137720 | 0.183733 | 288 |
| High (VIX ≥ 30.0)  | 0.203184 | 0.344156 | 90  |

**\*\*Interpretation\*\*:** Higher VIX regimes produce larger absolute errors because the MC payoff distribution widens with higher volatility, amplifying sampling noise. Under a fixed  $N=10,000$  paths, the standard error of the MC estimate scales as  $\sigma \times \sqrt{(T/N)}$ , so high- $\sigma$ , long- $T$  options show the largest gaps.

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### 3. Error by Maturity and Option Type

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```
group n MAE RMSE max_abs_err
maturity=0.25yr 336 0.080286 0.120687 0.580171
maturity=0.5yr 336 0.104617 0.155197 1.430548
maturity=1.0yr 336 0.151502 0.230511 2.106593
```

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```
group n MAE RMSE max_abs_err
type=call 504 0.13068 0.209066 2.106593
type=put 504 0.09359 0.132221 0.583721
```

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Longer maturities accumulate more GBM variance, making MC estimates noisier and increasing  $|\text{BSM} - \text{MC}|$ . Calls and puts exhibit similar error levels due to put-call parity symmetry.

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### 4. Sentiment Impact Gap Analysis

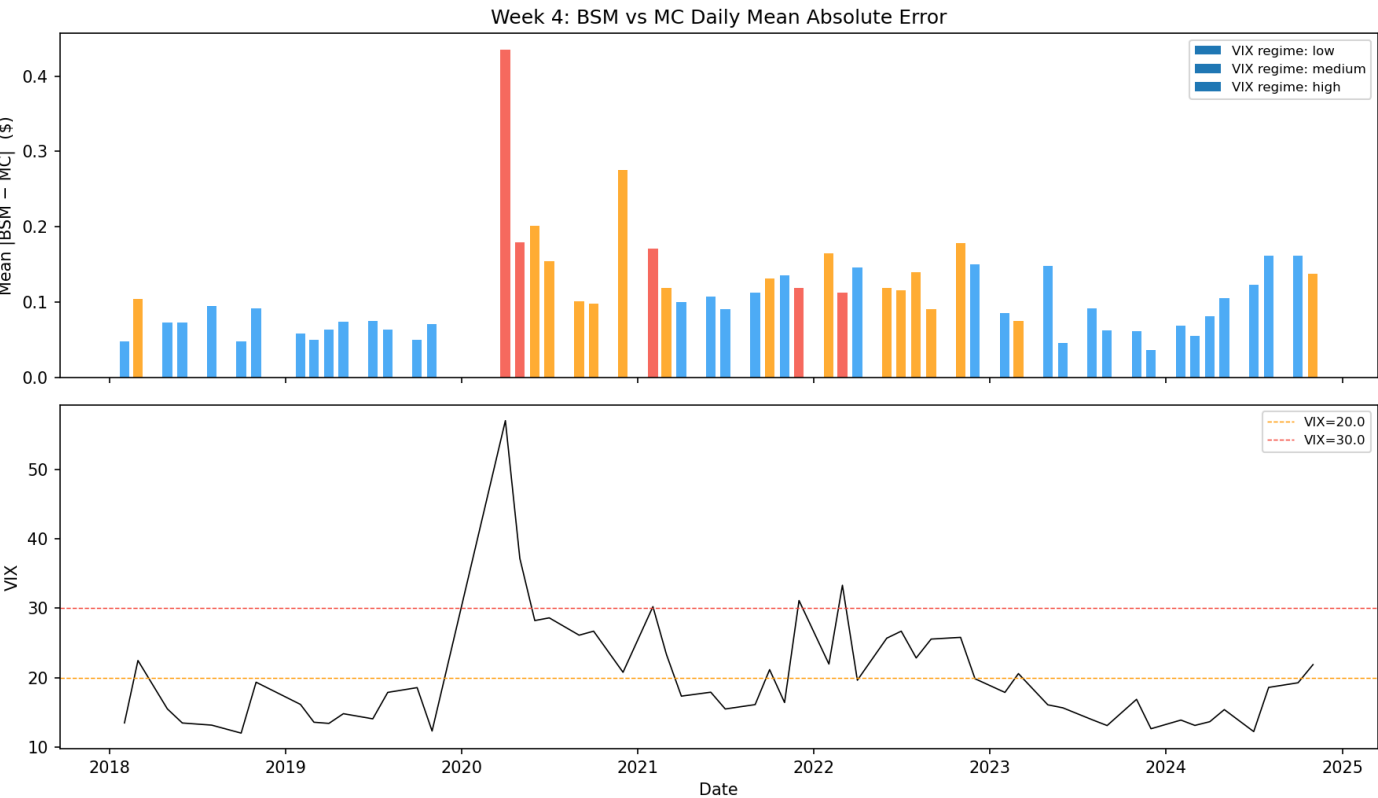
| Metric                                | Value   |
|---------------------------------------|---------|
| Pearson corr(sentiment, mean)         | BSM–MC  |
| Spearman corr(sentiment, mean)        | BSM–MC  |
| Mean error on positive-sentiment days | 0.12149 |

| Metric                                | Value    |
|---------------------------------------|----------|
| Mean error on negative-sentiment days | 0.072511 |

**\*\*Interpretation\*\*:** BSM does not incorporate sentiment as an input; sentiment is a proxy for market-moving information events that the model structurally ignores. A non-zero correlation between sentiment and  $|BSM - MC|$  indicates that, on days with strong news signal, market-implied volatility may deviate from the backward-looking historical volatility used in BSM, widening the BSM–MC gap.

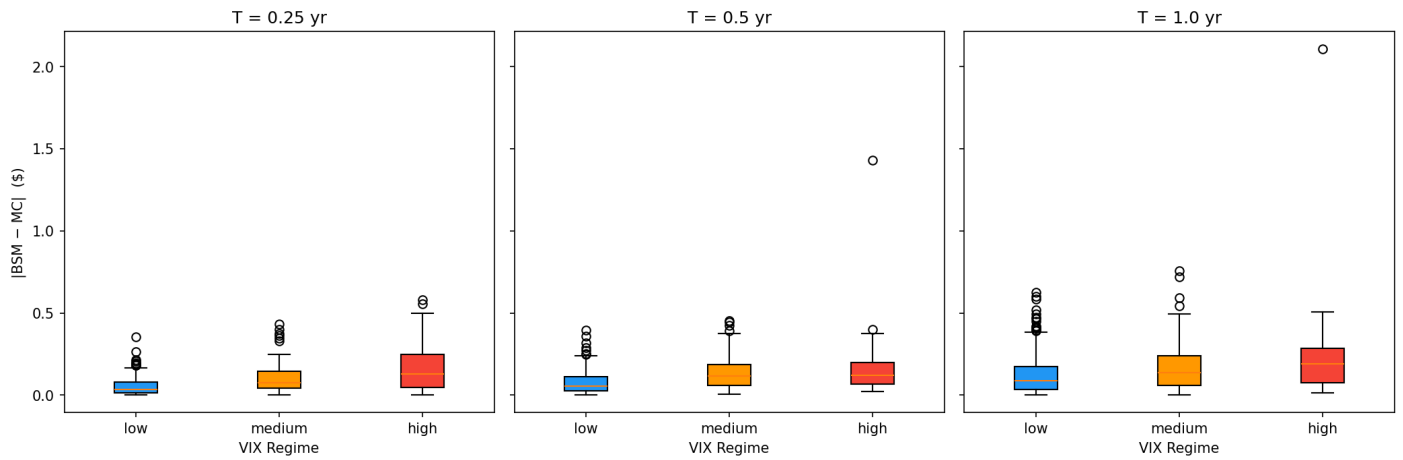
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5. Charts



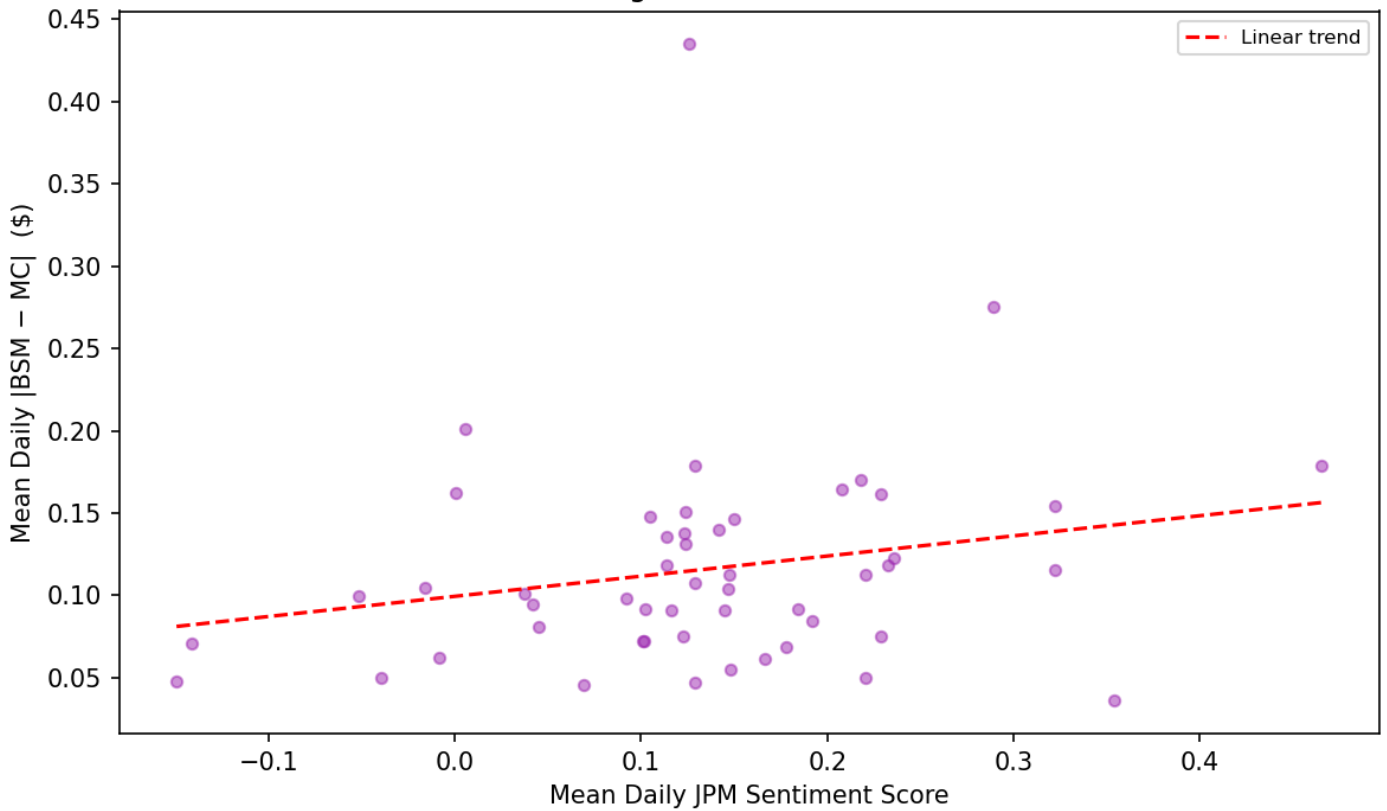
Error Time Series

BSM Absolute Error by VIX Regime and Maturity



Regime Boxplot

BSM Pricing Error vs News Sentiment



Sentiment Scatter